

Tracing and debugging

How each tool is switched on, what a run looks like, and what happens when the agent breaks.

Turning tracing on

This is the biggest difference, and it is visible in the code.

LangSmith: one environment variable

LANGSMITH TRACING=true

That is it. No imports, no code changes, nothing passed into the agent.

Proof is in the run list. The rows named `LangGraph` came from running the plain app:

```
uv run python -m src.agent "what's the weather in Chennai and what's 100/7?"
```

That file does not mention LangSmith anywhere. It still shows up with full latency, token and cost data.

Personal / Tracing / ilm-observability-lab

ilm-observability-lab ID Tracing Evaluators Automations Insights Engine **New**

Retention 14d Dashboard

Default View Last 1 day Threads **Traces** Runs Input Search input content

Add filter

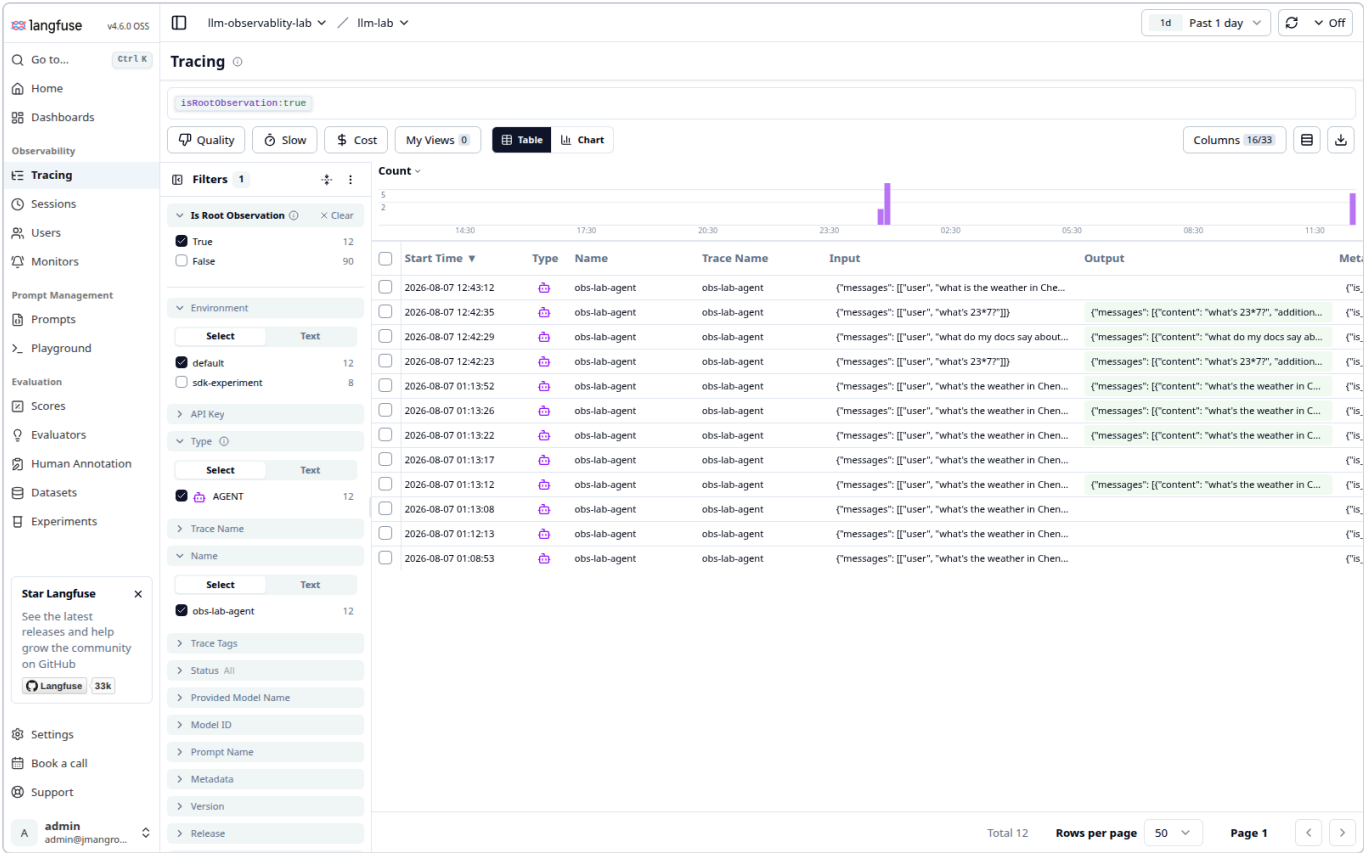
		Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation ...	Tokens	Cost	First Token	Tags
		forced-weather-failure	user: what is the weather...		RuntimeError("weather s...	8/7/2026, 11:51...	1.36s			279	\$0.0008175		
		LangGraph	user: what's the weather ...	ai: The weather in Chenn...		8/7/2026, 11:4...	2.13s			690	\$0.002295		
		obs-lab-tagged-run	user: what's the weather ...	ai: The weather in Chenn...		8/7/2026, 11:41...	2.77s			690	\$0.002295		phase3
		LangGraph	user: what do my docs s...	ai: Your docs say to retry ...		8/7/2026, 11:35...	3.96s			878	\$0.002795		
		obs-lab-tagged-run	user: what's the weather ...	ai: The weather in Chenn...		8/7/2026, 11:3...	2.18s			690	\$0.002295		manual-te
		ChatOpenAI	human: You are grading ...	ai: CORRECT The agent's...		8/7/2026, 11:17...	0.54s			130	\$0.0004675		
		LangGraph	user: what's 23*7?	ai: 23 * 7 equals 161.		8/7/2026, 11:17...	0.99s			577	\$0.001645		
		LangGraph	user: what's the weather ...		RuntimeError("weather s...	8/7/2026, 11:17...	0.63s			317	\$0.0011525		
		ChatOpenAI	human: You are grading ...	ai: CORRECT The agent's...		8/7/2026, 11:17...	0.62s			236	\$0.0008225		
		LangGraph	user: what do my docs s...	ai: Your docs say to retry ...		8/7/2026, 11:17...	3.26s			878	\$0.002795		
		LangGraph	user: weather in Mumbai,...		RuntimeError("weather s...	8/7/2026, 11:17...	1.41s			282	\$0.0008175		
		LangGraph	user: what's the weather ...		RuntimeError("weather s...	8/7/2026, 11:15...	0.66s			317	\$0.0011525		
		ChatOpenAI	human: You are grading ...	ai: CORRECT The agent's...		8/7/2026, 11:15...	0.56s			133	\$0.000475		
		LangGraph	user: what's 23*7?	ai: The result of 23 * 7 is 1...		8/7/2026, 11:15...	1.11s			581	\$0.001685		
		ChatOpenAI	human: You are grading ...	ai: First line: CORRECT Se...		8/7/2026, 11:15...	0.78s			288	\$0.0009975		
		LangGraph	user: what do my docs s...	ai: Your docs say: - Retrv ...		8/7/2026, 11:15...	14.07s			924	\$0.003255		

Look at the columns you get for free: Latency , Tokens , Cost , Error , Tags . The red rows are the weather tool failing. Nobody wrote error handling to make those appear.

Langfuse: one handler per call

```
handler = CallbackHandler()
graph.invoke(..., config={"callbacks": [handler]})
```

Nothing is recorded unless you pass that handler.



Which is better depends on what you care about. LangSmith gets you tracing in ten seconds, but it also captures everything in the process, including code you did not write. Langfuse costs you a line at every call site, and in exchange nothing leaves the process unless you asked for it.

One useful side effect: with both sets of keys in `.env`, a single run lands in **both** tools at once. Worth doing once.

What a single run looks like

Both give you the same core thing: a tree of what happened, with time and tokens on each node.

LangSmith

The screenshot displays the LangGraph web interface. On the left, a sidebar shows a list of runs under the 'llm-observability-lab' workspace. The main area is divided into two panels. The left panel, titled 'Trace', shows a hierarchical view of the execution steps, including 'agent', 'ChatOpenAI', 'should_continue', and 'tools'. The right panel, titled 'Attributes', shows a table of metadata for the selected run, including revision_id, LangSmith endpoint, project name, and runtime details.

Attribute	Value
revision_id	dc6b26a-dirty
LANGSMITH_ENDPOINT	https://api.smith.langchain.com
LANGSMITH_PROJECT	llm-observability-lab
LANGSMITH_TRACING	true
ls_integration	langgraph
ls_run_depth	0
langchain_core_version	15.3
langchain_version	1.3.14
library	langchain-core
library_version	1.5.3
platform	Linux-6.17.0-1030-oem-x86_64-with-glibc2.39
py_implementation	CPython
runtime	python
runtime_version	3.12.3
sdk	langsmith-py
sdk_version	0.10.16

2.13s , 690 tokens, \$0.0023 at the root. The graph:step and seq:step labels come from LangGraph and show the execution order.

Langfuse

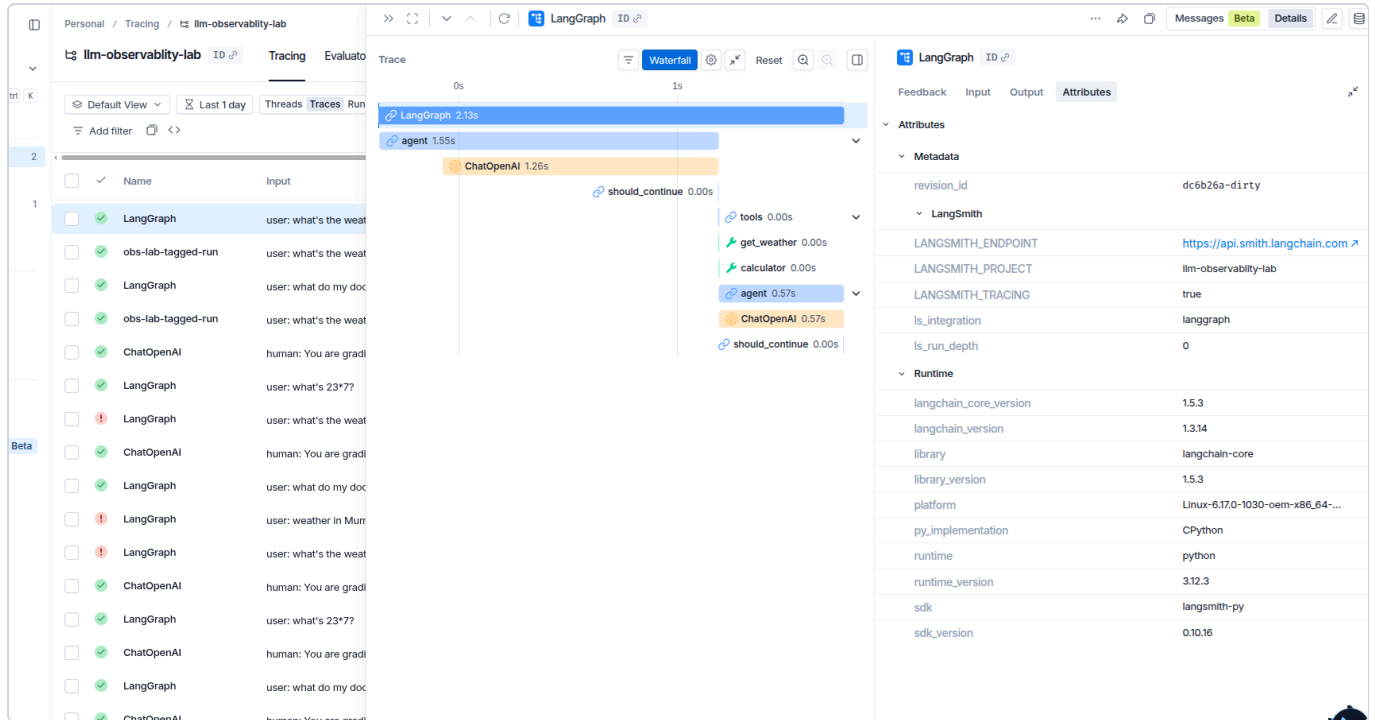
The screenshot displays the Langfuse web interface. On the left, a sidebar shows navigation options like 'Home', 'Dashboards', 'Observability', 'Tracing', 'Sessions', 'Users', 'Monitors', 'Prompt Management', 'Evaluation', and 'Experiments'. The main area is divided into two panels. The left panel, titled 'Trace', shows a hierarchical view of the execution steps, including 'agent', 'ChatOpenAI', 'should_continue', and 'tools'. The right panel, titled 'Preview', shows a table of input data, a 'Calculator' section, an 'Assistant' section, and a 'Corrected Output' section. Below the trace, there is a 'Graph' section showing a flow diagram of the agent loop.

Path	Value
is_langchain_root	true
ls_integration	"Langgraph"
scope.attributes.public_key	"pk-1f-492508f9-5e47-4c07-9ed4-2c7ba18d7126"
scope.version	"4.14.3"
scope.name	"Langfuse-sdk"
resourceAttributes.service.name	"unknown_service"
resourceAttributes.service.instance.id	"945f5e5f-ac70-4ac9-9704-b0b0c97bab12"
resourceAttributes.telemetry.sdk.version	"1.44.0"
resourceAttributes.telemetry.sdk.name	"opentelemetry"
resourceAttributes.telemetry.sdk.language	"python"

Same shape, plus a rendered graph of the agent loop underneath the tree.

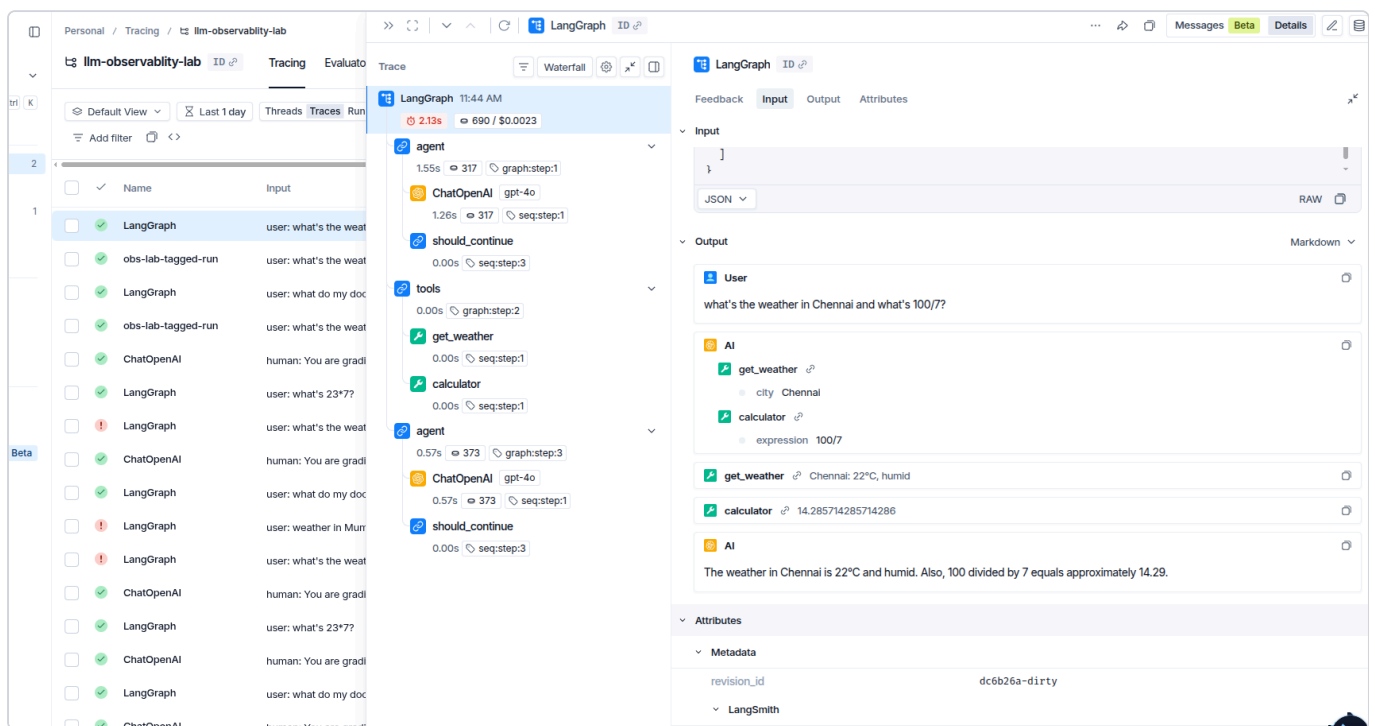
Timing view

LangSmith has a waterfall that makes it obvious where the time actually went:



The whole run is 2.13s and the `ChatOpenAI` call is 1.26s of it. Both tool calls are 0.00s. The model is the slow part, not your code. That is nearly always the answer, and it is worth seeing once.

The conversation itself

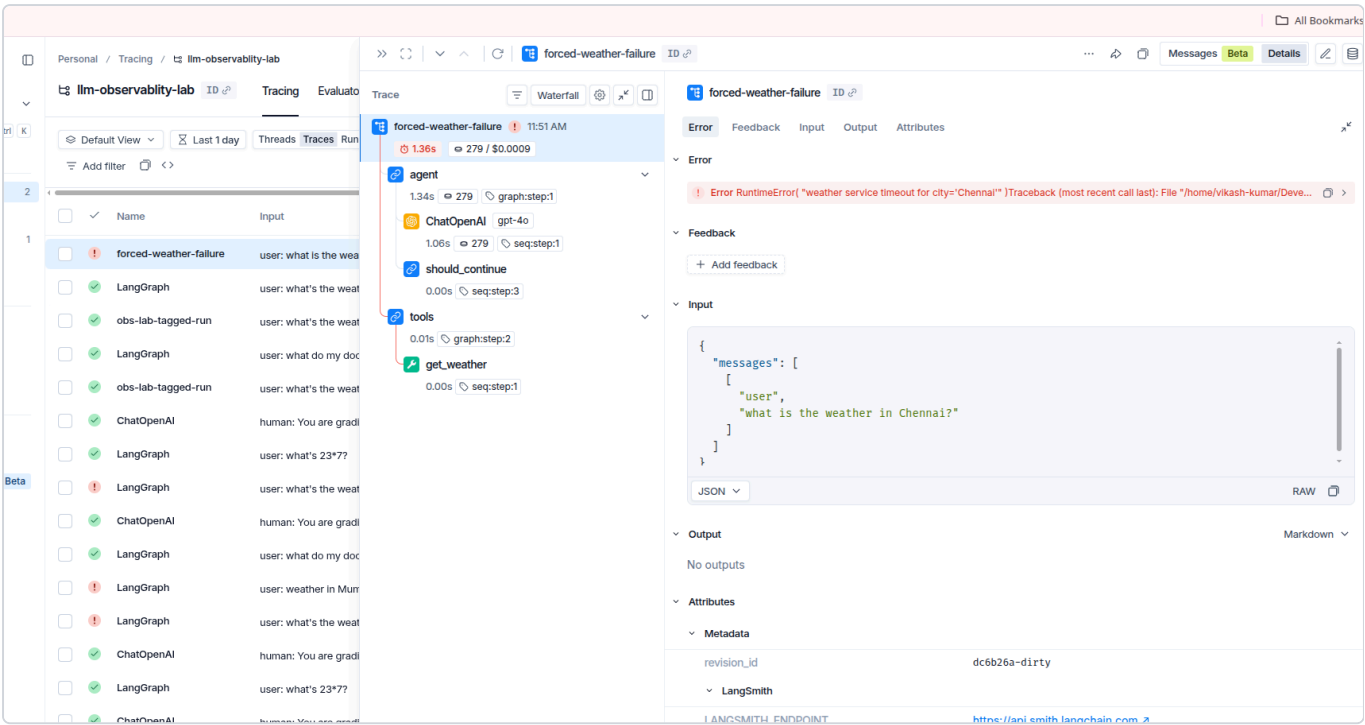


Both tools show the full message flow: what the model asked for, what each tool returned, what came back at the end.

When something breaks

This is where the two tools genuinely differ, and it matters more than any feature list.

LangSmith marks the root of the trace and puts the full Python traceback in a dedicated **Error** tab. Output says **No outputs**.



Langfuse marks every span the error passed through, all the way up.

The screenshot displays the Langfuse interface for a trace titled 'obs-lab-agent: 39d207c301916a86e0eb819c520508e2'. The left sidebar shows the 'Tracing' section. The main area shows the trace details, including the 'Preview' tab with input and assistant data. The trace tree on the left indicates an error in the 'obs-lab-agent' step, which is further detailed in the main panel.

Read the Langfuse tree carefully:

```
obs-lab-agent  ERROR
  agent
    ChatOpenAI      1.30s  270 -> 48  $0.001155
    should_continue
  tools            ERROR
    get_weather    ERROR  0.00s
    calculator      0.00s
```

`get_weather` failed, so `tools` is marked, so the whole trace is marked. `calculator` sits right next to it, untouched and green. The graph below even paints `__end__` red.

Verdict on errors: Langfuse wins. LangSmith tells you *that* the run failed and hands you a traceback. Langfuse shows you *where* in the agent it failed without you reading a single line of Python.

One detail worth knowing: `get_weather` shows `0.00s`. A real network timeout would take seconds. Zero means it failed on the first line of the function. That is how you tell a fake failure from a real one.

Grouping runs together

Both can thread separate runs into one conversation.

LangSmith calls them Threads:

The screenshot displays the LangSmith interface for a thread titled 'llm-observability-lab'. The left sidebar contains navigation options like 'Experiments', 'Queues', and 'Shortcuts'. The main area shows a list of messages and their corresponding AI responses. The 'INPUTS' section shows a user message: 'what's the weather in Chennai and what's 100/7?'. The 'OUTPUTS' section shows the AI's response, which includes a weather check and a calculation. The right sidebar shows 'Stats' for the thread, including 'First start', 'Last end', 'Latency', and 'Total cost breakdown'.

Langfuse calls them Sessions:

The screenshot displays the Langfuse interface for a session titled 'llm-observability-lab'. The left sidebar contains navigation options like 'Home', 'Dashboards', 'Observability', 'Sessions', 'Users', 'Monitors', 'Prompt Management', 'Evaluation', 'Human Annotation', 'Datasets', and 'Experiments'. The main area shows a table of sessions. The 'Filters' section on the left allows filtering by 'Environment' (default, sdk-experiment), 'Session ID', 'User IDs', 'Trace Tags', 'Metadata', 'Bookmarked', 'Session Duration', 'Traces Count', 'Input Tokens', 'Output Tokens', 'Total Tokens', 'Input Cost', 'Output Cost', 'Total Cost', 'Categorical Scores', 'Numeric Scores', 'Boolean Scores', 'Comment Count', and 'Comment Content'. The table shows one session: 'session-001', created at '2026-08-07 01:08:53', with a duration of '11h 34m 20s', environment 'default', user 'vikash', and 12 traces. The bottom right shows 'Rows per page' set to 50, 'Page 1 of 1'.

One session, session-001 , holding 12 traces over 11 hours, attributed to user vikash .

That 11 hour duration is a warning, not a feature. The session ID is hardcoded in [traced_run.py](#), so every run ever executed piles into the same session. In a real app you generate one per conversation.

The two tools name these fields differently:

	LangSmith	Langfuse
Tags	<code>config["tags"]</code>	<code>metadata["langfuse_tags"]</code>
User	<code>metadata["user_id"]</code> , any key you like	<code>metadata["langfuse_user_id"]</code> , reserved name
Session	<code>metadata["session_id"]</code> , any key you like	<code>metadata["langfuse_session_id"]</code> , reserved name

LangSmith lets you invent your own keys and filter on them. Langfuse only promotes specific `langfuse_` - prefixed keys into real UI features, and treats everything else as plain metadata. You trade magic strings for purpose-built screens.
